9) $y = \frac{\sqrt{2x^3 + 3}}{(x^4 - 3)^3}$

10) $y = (2x^2 - 5)^3 \sqrt{x^2 - 2}$

Use logarithmic differentiation to differentiate each function with respect to *x*. You do not need to simplify or substitute for *y*.

11) $y = \frac{(5x - 4)^4}{(3x^2 + 5)^5 \cdot (5x^4 - 3)^3}$

12) $y = (x + 2)^4 \cdot (2x - 5)^2 \cdot (5x + 1)^3$

13) $y = (5x^5 + 2)^2 \cdot (3x^3 - 1)^3 \cdot (3x - 1)^4$

14) $y = \frac{(x^2 + 3)^4}{(5x^5 - 2)^5 \cdot (3x^2 - 5)^2}$

15) $y = (3x^3 - 4)^5 \cdot (3x - 1)^3 \cdot (5x^3 - 2)^2 \cdot (x + 3)^4$

16) $y = \frac{(4x^2 - 5)^2}{(2x - 3)^4 \cdot (5x^4 - 2)^5 \cdot (3x^2 - 4)^3}$